The Fourth Horn: Local Pais–Uhlenbeck Modified Inertia Is Excluded by an Exceptional-Point Cap and a Frequency-Selection No-Go

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Abstract

Scale Without Law (10.5281/zenodo.21016309) closed three routes to a covariant modified-inertia (MI) MOND: local higher-derivative actions (Ostrogradsky ghost), field-theoretic completions (Cassini), and nonlocal kernels over passive baths (the anti-MOND sign, sharpened to a state clause in 10.5281/zenodo.21139029). One route remained unwalked: a **local, preferred-frame, PT-quantized** higher-derivative worldline theory — motivated by the facts that a preferred frame voids the Galilei premise of Milgrom's 1994 nonlocality theorem, and that Bender–Mannheim PT quantization dissolves the free Pais–Uhlenbeck (PU) ghost. This note walks that route end-to-end, adversarially verified, and closes it — constructively, with theorems rather than estimates, and framed generically: the results apply to *any* MOND-type modified inertia, not only the framework variant used for normalization. **(G0)** No local worldline Lagrangian $L(x, \dot{x}, \ddot{x})$ has the algebraic MI law $F = m\mu(|a|/a_0)a$ as its exact Euler–Lagrange equation, preferred frame or not: acceleration-dependence forces a genuine fourth-order (PU-class) theory, so the honest target is an *effective* μ. **(The exact identity.)** For the general PU-class action with nonlinearity $k(\ddot{x}^2/a_0^2)$, circular orbits obey $a[1 - (\Omega/w\_eff)^2] = g\_N$ with $w\_eff^2 = w^2/k'$ — the effective inertia is **a frequency law, $\mu\_eff = 1 - (\Omega/w\_eff)^2$, never an acceleration law**. At fixed g_bar it depends on orbital frequency, i.e. on velocity and radius separately: a single 200⟶300 km s^{-1} comparison at fixed g_bar shifts the prediction by 0.176 dex, exceeding the *entire* 0.108 dex observed RAR scatter, and dwarf-regime parameters admit no circular solution at all. This universality kill is a classical identity, independent of quantization or inner product. **(The sign converse.)** The softening (MOND) direction obtains *if and only if* the $\ddot{x}^2$ term carries the ghost sign — the local cousin of the nonlocal sign theorem: in local higher-derivative inertia, the MOND direction and the ghost are inseparable. **(The exceptional-point cap.)** PT quantization rescues reality only in the unbroken phase, and the unbroken condition $w\_eff \geq 2\Omega_0$ (harmonic-exact) caps the healthy-phase softening at **$\mu\_eff \geq 1/2$**, with the boundary a Jordan-block exceptional point at which deeper-MOND circular orbits *cease to exist* (non-existence, not runaway, is the operative kill); on flat-rotation-curve anchoring the cap is plausibly $\mu\_eff \gtrsim 0.854$, pending a Coriolis-coupled linearization we state as the named residual calculation. Deep MOND ($\mu \longrightarrow 0$) is therefore unreachable inside the healthy phase regardless. **(Scales.)** Independently, no constant PU scale w works: hidden modes (w above laboratory/ephemeris floors) render the mechanism galaxy-inert, while MOND-tuned w survives Solar-System bounds only through its external-field shielding and is then excluded by wide binaries at the ~10^5 level — the frequency-selection failure restated in data. The trichotomy is thus a **quadrichotomy**, and its four horns now converge on a single statement: an acceleration-selected μ requires the effective stiffness w_eff to know the *frequency content of the trajectory* — a nonlocal-in-time response — whose only sign-viable realization is the pumped anharmonic

medium specified by the state clause. The unwritten MI completion is no longer an open field; it is a single, sharply specified, presently unoccupied requirement, reached independently from four directions.

1. Why this route deserved a walk

Milgrom (1994) proved that a Galilei-invariant MI action with the correct limits must be strongly nonlocal; the framework at hand breaks boost invariance by construction (its preferred frame is the source of its published s̄^TX prediction), voiding that theorem's premise. And the classical obstruction to local higher-derivative dynamics — the Ostrogradsky ghost — is known to dissolve for the free PU oscillator under PT quantization (Bender & Mannheim 2008): the ghost's negative classical energy (verified symbolically: $E = -mA^2w^2/2$ for the fast mode) coexists with a real, positive, unitary quantum spectrum under the CPT inner product. Local + preferred-frame + PT was therefore the one combination not covered by the banked no-gos: the sole surviving hope for a covariant MI needing no bath, no pump, and no medium.

2. G0: the exact law is non-Lagrangian-local

First, a structural theorem (committed as `reviews/pt_worldline_g0_theorem.py`, sympy-verified): for any local $L(x, \dot{x}, \ddot{x})$, either $\partial^2L/\partial\ddot{x}^2 \neq 0$ — in which case the Euler–Lagrange equation contains x'''' and the theory is genuinely fourth-order — or L is linear in $\ddot{x}$, in which case the acceleration term is exactly a total derivative plus an $L(x, \dot{x})$ piece, and the surviving inertia $\partial^2L/\partial\dot{x}^2$ can depend on position and velocity but never on acceleration. No local Lagrangian has $F = m\mu(|a|/a_0)a$ as its exact equation of motion; the preferred frame removes Milgrom's obstruction only to expose a sharper, purely variational one. The target of the remaining gates is therefore an *effective* μ emerging from the PU tower.

3. The exact identity, the sign converse, and the universality kill

Take the general PU-class worldline theory in the preferred frame, $L = (m/2)\dot{x}^2 - (m\, a_0^2/2w^2)\, k(|\ddot{x}|^2/a_0^2) - m\Phi$, with k an arbitrary smooth nonlinearity. On circular orbits an exact identity holds (no adiabatic approximation; derived twice by independent routes):

$$a\cdot[1 - (\Omega/w\_eff)^2] = g\_N,\quad w\_eff^2 \equiv w^2/k'(y),\quad y = a^2/a_0^2 \text{ — i.e. } \boldsymbol{\mu\_eff = 1 - (\Omega/w\_eff)^2}.$$

Three theorems follow. **(i) Frequency, not acceleration.** μ_eff is selected by the orbital *frequency* Ω relative to the (possibly y-dependent) stiffness — not by a/a_0. Demanding that it mimic $\mu(a/a_0)$ forces k′ to take different values at the same y for different radii (computed: a factor R_2/R_1, e.g. 0.074 vs 0.74 between 2 and 20 kpc) — k′ would have to not be a function, which is the G0 obstruction re-derived inside the PU frame. Observationally: at fixed g_bar = $0.7a_0$, moving from a 200 to a 300 km s^{-1} galaxy shifts μ_eff by 0.176 dex — more than the entire observed RAR scatter (0.108 dex) from one comparison — and dwarf parameters (30–80 km s^{-1}) yield μ_eff < 0: no circular solution exists. Nature has sampled (low-a, high-Ω) and (low-a, low-Ω) and drawn one curve; this class cannot. The kill is a classical identity, untouched by any choice of quantization or inner product. **(ii) The sign converse.** Softening (μ_eff < 1, the MOND direction) obtains for k′ > 0, which is precisely the *ghost sign* of the $\ddot{x}^2$ term; the healthy sign stiffens (anti-MOND). Verified with its converse: in local higher-derivative inertia, **the MOND direction and the ghost are inseparable** — the local counterpart of the nonlocal sign theorem, now closing the sign question on both sides of the locality divide. **(iii) A tail curiosity.** The unique exponent matching the framework's Newtonian tail (k′ ~ y^(−1/2)) collapses the force to a pure 1/R form (the Grumiller Rindler-force profile) with the longitudinal stiffness κ_‖ = k′ + 2yk″ vanishing identically — a marginal mode, not a MOND law.

4. The exceptional-point cap

PT quantization cures the PU ghost only in the *unbroken* phase. For a particle in a well, the two PU frequencies are real iff w_eff ≥ $2\Omega_0$; at equality they merge in a Jordan block (an exceptional point), beyond which the classical frequencies are complex conjugates and the PT symmetry of the state is spontaneously broken — an instability no inner product repairs (verified numerically: growth rate Re λ = 0.30 Ω_0 at w = 1.8 Ω_0, matching the root computation, and confirmed against long orbit integrations both sides of threshold). Combining with the identity, the unbroken phase satisfies (harmonic-exact, orbit branch):

> **unbroken PT ⍰ μ_eff ≥ 1/2** — the healthy phase caps softening at 50%, with the boundary at g_bar ≈ (2/3)a_0 under harmonic anchoring; deeper softening fails not by runaway but because the circular orbit *ceases to exist* at the exceptional point.

For flat-rotation-curve anchoring the cap tightens to μ_eff ≳ 0.854 (≈15% softening) up to a Coriolis-coupled O(1) factor: the full in-plane linearization (an eighth-degree characteristic polynomial in κ_‖, k′) is the one named residual calculation, and we do not claim the flat-curve constant as exact. What needs no residual calculation: deep MOND (μ_eff ⟶ 0, the regime that carries all of MOND phenomenology) lies unreachable inside the healthy phase under *any* anchoring, and the universality kill of §3 operates at all μ_eff regardless. A flagged quantum risk is recorded without being leaned on: the induced quartic carries a negative ghost-momentum self-coupling (a Smilga-malicious candidate) whose nonperturbative benignity is undetermined, and the perturbative reality argument covers the saturating nonlinearity only.

5. No scale works

The constraint chain over the PU scale w admits no window: hidden modes (w above the ephemeris/pulsar/interferometer/atomic floors, up to w ≳ 10^{21} s^{-1}) make the galactic effect ~10^{-39} — inert; a MOND-tuned w ≈ 140 H_0 genuinely evades LISA-Pathfinder, torsion-balance and LIGO bounds through its built-in external-field shielding (an honest pass, recorded), survives Saturn only marginally, and is then excluded by **wide binaries at the ~10^5 level** — systems at galactic y but a thousand times the orbital frequency, which is the frequency-selection failure of §3 expressed in data. A monomial interpolation between the regimes is impossible (the required exponent is negative where stability requires positive), and a designer non-monotone stiffness would imprint ~15 dex of scatter on the RAR.

6. The quadrichotomy, and the convergence

The no-go structure of covariant modified inertia now reads: **(I)** local higher-derivative, Dirac-Hermitian — Ostrogradsky ghost; **(II)** field-theoretic — Cassini slip; **(III)** nonlocal over passive baths — anti-MOND sign (a state clause; escape requires a pumped anharmonic medium that free fields and the de Sitter horizon provably cannot supply); **(IV)** local higher-derivative, preferred-frame, PT-quantized — the exceptional-point cap, the sign−ghost inseparability, and the frequency-selection no-go (this note). All four horns converge on one sentence: *an acceleration-selected μ requires the inertial response to know the frequency content of the whole trajectory* — a nonlocal-in-time kernel — *and the only sign-viable source for such a kernel is an inverted (non-equilibrium) medium* per the state clause. The unwritten MI completion is thereby specified from four independent directions and occupied by no known physics. The constructive residue of this note: any Lagrangian PU realization is viable only as a MOND-inert ultraviolet spectator ($w \gtrsim 10^4\ s^{-1}$, PT-curable per Bender–Mannheim), and the named next calculation is either the flat-curve exceptional-point closure or the direct construction and testing of the nonlocal kernel $w\_eff^2[x(\cdot)]$ against causality, stability, and the wide-binary bound.

Scope and credit

The PU oscillator is Pais & Uhlenbeck (1950); the PT rescue of its free spectrum is Bender & Mannheim (2008); the benign/malicious ghost distinction is Smilga; the nonlocality theorem whose premise the preferred frame voids is Milgrom (1994); the 1/R force profile is Grumiller (2010); the wide-binary sensitivity anchors on the published analyses (Banik et al. 2024; Chae 2023-25 — the kill margin dwarfs their disagreement). The results are MOND-generic; the framework enters only through the normalization $a_0 = c^2\sqrt{(\Lambda/32\pi)}$ and its preferred frame, and its earlier over-claims were publicly retracted (2026-06-23) — this note claims a closure, not a completion. Adversarially verified end-to-end, including one corrected constant (the cap is $\mu \geq 1/2$ on the orbit branch, not 3/4 as first computed) and a re-graded flat-curve statement; the gauntlet and 22-check verifier scripts are committed alongside.

References

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Reproducible: `reviews/pt_worldline_g0_theorem.py` *(G0),* `reviews/pt_gates/{pu_g2_pt_spectrum, pu_g3_constraint_chain, pu_g4_effective_mu}.py` *(gates),* `reviews/pt_gates_verify_2026-07.py` *(22-check adversarial verifier) — all exit 0. Companion:* `PT_GATES_VERDICT_2026-07.md`*. The author retracted all earlier "theory of everything" claims (2026-06-23); this note makes none.*